MBAN 5540: Prescriptive Analytics & Optimization
Instructor: Majid Taghavi

Assignment 2

Due by: Thursday, February 05, 2026, at 10:00 a.m.

Please submit at most five files through **Brightspace**: One PDF file for the solution to Question 1. Two notebooks and two .lp files for the primal and dual models in Question 2.

Question 1

RRB specializes in small watercraft, including canoes, kayaks, and dinghies. These are made using two processes. The kayaks are made from polyethylene plastic heated in molds, whereas the other products are made from layers of fiberglass or Kevlar built over a mold. In all cases, after the basic watercraft shape is made, it requires some further cutting and finishing as well as the installation of fittings, inspection, and packaging. RBB formulated their problem as a linear program as follows:

- C_1: # of of small canoes
- C_2: # of of regular canoes
- K_1: # of one-person kayaks
- K_2: # of two-person kayaks
- D: # of dinghies

The LP Model:

$$
\begin{array}{rll}
\max & 32C_1 + 41C_2 + 42K_1 + 55K_2 + 25D & \\
\text{s.t.} & 2.0K_1 + 2.1K_2 \leq 120 \text{ hours} & \text{(Kayak mold prep time)} \\
& 1.8K_1 + 2.3K_2 \leq 240 \text{ hours} & \text{(Kayak oven time)} \\
& 2.2C_1 + 2.5C_2 + 1.6D \leq 210 \text{ hours} & \text{(Kevlar/fiberglass time)} \\
& 1.5C_1 + 1.5C_2 + 1.8K_1 + 2.0K_2 + 0.9D \leq 320 \text{ hours} & \text{(Cutting/finishing time)} \\
& 1.5C_1 + 2.0C_2 + 2.2K_1 + 2.7K_2 + 1.4D \leq 320 \text{ hours} & \text{(Fitting installation time)} \\
& C_1 + C_2 \geq 50 \text{ units} & \text{(Minimum Canoe output)} \\
& K_1 + K_2 \geq 50 \text{ units} & \text{(Minimum Kayak output)} \\
& D \geq 10 \text{ units} & \text{(Minimum Dinghy output)} \\
& C_1, C_2, K_1, K_2, D \geq 0 & \text{(non-negativity)}
\end{array}
$$

After solving the model, the following sensitivity report is available.

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
R4C2	# made Small Canoe	13.4065934	0	32	4.08	1.2162963
R4C3	# made Regular Canoe	65.8021978	0	41	1.38215488	2.98529412
R4C4	# made Single Kayak	0	-4.55238095	42	4.55238095	1E+30
R4C5	# made Double Kayak	57.1428571	0	55	1E+30	4.78
R4C6	# made Dinghy	10	0	25	3.12307692	1E+30

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
R11C7	Min Canoe Total	79.2087912	0	50	29.2087912	1E+30
R12C7	Min Kayak Total	57.1428571	0	50	7.14285714	1E+30
R13C7	Min Dinghy Total	10	-3.12307692	10	49.962406	10
R6C7	Kayak mold Total	120	6.01465201	120	15.1212121	2.71111111
R7C7	Kayak oven Total	131.428571	0	240	1E+30	108.571429
R8C7	Kelvar Total	210	3.84615385	210	28.5142857	4.35714286
R9C7	Finishing Total	242.098901	0	320	1E+30	77.9010989
R10C7	Fittings Total	320	15.6923077	320	3.48571429	19.4415584

Answer Report 1 | Sensitivity Report 1 | Sheet1

Use the report (not Excel solver) to answer the following questions and justify your answer:

1. What is the production plan and total profit? Assume that fractional output is acceptable
2. Would the solution change if the minimum number of canoes made was increased to 60? If the minimum number of kayaks was increased to 60? If the minimum number of dinghies was increased to 20?
3. It has been determined that the oven used to heat the kayak mold must be shut down in one month, leaving only 150 hours of time. Will this affect the solution or the profit?
4. Extra fitting time can be obtained by hiring someone at a minimum wage of $10.85. Should we hire someone? If so, for how many hours?
5. Suppose they can also get someone to help finish at $12 per hour. Should we hire someone? If so, for how many hours?
6. Management has noted that the plan calls for producing all products except the single kayak. What would be the impact on profits if they decided to make a single kayak?
7. Dinghies are only being made because they have to deliver 10 of them. What would the profit per dinghy need to be to make us build them, given that they increase total profits?
8. RBB has not made racing kayaks for many years, but still has the mold to produce them. To make a racing kayak, they would need 2.4 hours of mold prep time and 2.1 hours of mold oven time. They would also need 2.1 hours of cutting/finishing time and 2.5 hours for installation of fittings. How much profit would a racing kayak need to generate to make it profitable enough to consider?

Question 2

An LP model is expressed in terms of its technology matrix (A), RHS vector(b), and cost vector (c) in an Excel file named 'Params.xlsx' with 3 sheets (attached). We would like to minimize the objective function. Write a Python code (or any other programming language) in a Jupyter notebook that does the following steps:

1. Read each sheet of the Excel file and create A, b, and c.
2. Define a function called 'Min_Vector'. This function declares the model, adds all the decision variables and constraints, sets the objective, and writes the model to a file called 'Primal.lp'. Name your variables x in this function.
3. Solve the model.
4. Define another function called 'print_solution' which takes a model as input, and prints all the decision variables that have a positive optimal value next to their names. It also prints the optimal value of the objective function.
5. Define another function called 'Max_Vector' with all the functionalities of the 'Min_Vector' function. This 'Max_Vector' function should solve the dual of the original problem defined by A, b, and c. Name your variables π in this function and write the model into a file called 'Dual.lp'.
6. Solve the dual model.
7. Use the 'print_solution' function that you previously defined to print the solutions of the dual model. Verify that the optimal objective function value is the same for the primal and dual models.

Use Markdown cells to put some explanation before every code cell that you use in a Jupyter notebook.

Good Luck!